

AMA1D03 Introduction to Pension Mathematics

Case Study and Presentation

The case study has 2 sections: Section I is compulsory and Section II is elective. Assessments include Oral presentation and final type-written report. In section I, you are given the background information of the case. Based on the current situation in Hong Kong, recommend some methods for the pension and support your arguments with evidence. In section II, select **ONE** of the topics (Case I/II/III) or other topics related to Pension Mathematics.

General requirements of case study:

1. Each group has at most 2 members.
2. Division and contribution of works in a group should be specified in the appendix of the final report, if the group has more than one member.
3. Number of pages: minimum 6 pages and maximum 12 pages (e.g., max 2 members x max 6 pages per member) (covering page and appendix are not included).
4. Margin (top, bottom, left, right) and spacing: 1" and 1 line.
5. Font and size: Times New Roman and 12.
6. Submission of soft copy checked by Turnitin.
7. Oral presentation will be replaced by Video presentation (max. 10 minutes, 5 minutes per member x max 2 members). Upload Video via blackboard before due time.
8. Type-written report in pdf file should be submitted to complete the tasks.
9. Both video and report must be submitted via **Blackboard** on or before due time **5 July 2026 23:59**.

Section I

A 55-year-old single woman expect to retire at age 65. Her job is stable and her current salary is \$30,000. She is now healthy and does not have medical insurance. She has savings around 1 million dollars. Her MPF account is valued around 1 million dollars.

She is living in her property with outstanding loan balance of \$900,000 in Tsuen Wan and her current monthly repayment of \$16,000. Following the repayment schedule, she expects to settle the property loan in 10 years. The size of her flat is 480 square feet with 2 bedrooms and valued around 4.9 million dollars and the price of her property is expected to remain unchanged in 10 years. The market rent of her property is around \$14,000 per month. The age of her property is around 35-year-old.

Section II

Case I

The base level of protection against financial security is social insurance established by the government. This may cover income benefits at retirement, death and disability benefits, medical expense, and worker compensation. The employer-provided benefits make up the second layer of coverage or protection. Many employers also have private pension and retirement plan to provide income for employee after retirement. The third layer of retirement income is personal savings and income from family members.

Discuss the **pension system** of **Two** developed/developing countries/regions (Two countries/regions must be in different continents), including the following contents:

- (a) Descriptions of the **pension system**. For example, defined contribution or defined benefit, public or private or both, contributions, benefits, when and how to get benefit, or any specific features.
- (b) Compare and Analyze the **pension system** of this country: e.g. the advantages and disadvantage, contributions and benefits, demography, inflation, average life, percentage of retirees...etc.
- (c) How can we support our retirement under this **pension system**, by using the social security system and private pension system? Illustrate the application of this **pension system** by giving examples by giving examples.

Case II

Any person in Hong Kong aged 65 or above is eligible for Normal old age living allowance/Higher old age living allowance if the income and asset not exceeding the prescribed limits. In fact, "Income" includes wages from employment; income from handiwork, business, etc. (including salaries, wages, monthly commissions or bonuses received, and monthly income from self-employment); retirement benefits/pensions; and net income on rentals collected. Contributions from family members, relatives or friends, and monthly payments received under reverse mortgage schemes are excluded, but any unspent and accumulated amount of savings/cash in hand generated will be treated as "assets".

Discuss the **reverse mortgage** in **Two** developed/developing countries/regions (One of the regions must be Hong Kong), including the following contents:

- (a) Descriptions of the **reverse mortgage** and its application in pension system. For example, public or private or both, contributions, benefits, when and how to get benefit, or any specific features.
- (b) Compare and Analyze the **reverse mortgage** and illustrate by examples: e.g., the advantages and disadvantage, contributions and benefits, any other alternatives...etc.
- (c) How can we use the reverse mortgage to support our retirement together with the social security system? Illustrate the application of this **reverse mortgage** by giving examples.

Case III

Mandatory Provident Fund (MPF) launched since 1st Dec 2000. MPF, a defined contribution scheme for the employee in Hong Kong, is one of the pillars supporting retirement living. Suppose two employees Peter and Mary with monthly contribution of \$1,000 in a defined contribution plan which is 5% of their salaries \$20,000 when their ages are both 20 and 40, respectively. Their employers also contribute the same amount. Peter's contribution is invested in Global fund and Mary's contribution in Hong Kong fund since 1st Dec 2000. Suppose their salaries are adjusted based on the Consumer Price Index, CPI in Hong Kong due to inflation. They are thinking about whether to add extra contribution to their pension funds, if they plan to retire at their age 65.

Discuss the **Defined Contribution** Plan, including the following contents:

- (a) Descriptions of the **Defined Contribution** Plan and its application in pension system. For example, types of schemes, Mandatory Pension Fund (MPF), mandatory or voluntary or both, contributions, benefits, when and how to get benefit, investment and returns, tax-deductible or any specific features.
- (b) Compare and Analyze the **Defined Contribution** Plan and illustrate by examples: e.g., the advantages and disadvantage, contributions, and benefits, how much living expense covered, replacement ratio, inflation, average investment returns, and expense of funds...etc.
- (c) How can we support our retirement under this **Defined Contribution** Plan with the social security system? Illustrate the application of this Defined Contribution Plan by giving examples by giving examples.

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Some reference links about Pension and Social security in Hong Kong (for reference)

https://www.swd.gov.hk/oala/index_e.html#s4

https://mfp.mpfa.org.hk/eng/mpp_chart.jsp

<https://www.hkmca.hk/eng/>

https://www.hkmc.com.hk/eng/our_business/reverse_mortgage_programme.html

https://www.hkmc.com.hk/eng/our_business/policy_reverse_mortgage_programme.html

<https://www.investopedia.com/terms/i/ira.asp>

<https://www.investopedia.com/terms/p/pay-as-you-go-pension-plan.asp>

<https://www.investopedia.com/terms/c/cpp.asp>

<https://www.labour.gov.hk/eng/public/wcp/ConciseGuide/11.pdf>

<https://finance.yahoo.com/quote/2800.HK/?p=2800.HK>